Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Somatic Mutations in Muscle and Bone Exposed to Simulated Space Radiation and Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hjdonahue@vcu.edu	**Fax:**	FY 804-827-0290
PI Organization Type:	UNIVERSITY	**Phone:**	804-828-7263
Organization Name:	Virginia Commonwealth University		
PI Address 1:	Department of Biomedical Engineering		
PI Address 2:	401 West Main Street		
PI Web Page:			
City:	Richmond	**State:**	VA
Zip Code:	23284-3067	**Congressional District:**	3
Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	06/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance per NSBRI (formerly 5/1/15-4/30/17)--Ed., 7/7/15		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Imamura, Yuka Ph.D. (The Pennsylvania State University) Makova, Kateryna Ph.D. (The Pennsylvania State University) Eckert, Kristin Ph.D. (The Pennsylvania State University) Lang, Charles Ph.D. (The Pennsylvania State University)		
Grant/Contract No.:	NCC 9-58-MA04201		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Specific Aim: Characterize somatic genetic variation in nuclear and mitochondrial DNA, and resulting changes in RNA, from skeletal muscle and bone in male mice undergoing simulated microgravity (hind limb unloading, HLS) in the presence and absence of radiation from high charge and energy (HZE) particles, specifically those from 16O and protons. Aims will addressed following hypotheses: * Mutational signatures in bone and muscle will be altered in mice exposed to HZE and proton radiation during a period of HLS. * HZE and proton radiation will induce somatic mutations in both nuclear and mitochondrial DNA. * Accumulation of somatic mutations induced by the interaction of radiation and HLS in mitochondrial DNA will be greater than in nuclear DNA. * Somatic variation produced by radiation and HLS is maladaptive, enhances the catabolic state, and accelerates the development of functional decline in both bone (osteopenia) and muscle (sarcopenia). Key findings: Several events impacted progress on the project. First, while the grant was awarded 6/1/2015 we did not receive Institutional Animal Care and Use Committee (IACUC) approval until 8/18/2015. Second, Dr. Donahue moved his lab from Penn State to Virginia Commonwealth University on December 31, 2015 and the grant did not transfer until 1/1/2017. The third reason, and by far the most important, has to do with the nature of our specific aim. The project has one specific aim--exposing mice to simulated microgravity and, at the same time, a very specific simulated space radiation. There is only one opportunity for exposure to the simulated space radiation and that can only happen at the National Space Radiation Laboratory operated by Brookhaven National Lab on Long Island, NY. The earliest Dr. Donahue was able to reserve beam time to expose mice to radiation was June 8, 2016. Thus, all the work reported was completed between June 8, 2016 and 5/31/2017. We successfully exposed over 90 mice to simulated microgravity and space radiation (this was accomplished May 27 to June 17, 2016, see task progress section). • Mice were exposed to hind limb unloading starting on May 1, 2016. Mice were exposed to simulated space radiation while undergoing hind limb unloading on June 2, 2016. Hind limb unloading continued until June 21, 2016 at which time mice are sacrificed and sample collected. • Bioinformatics analyses for somatic mutations is now ongoing. • Completed adjunct study examining the effect of mechanical loading on radiation-induced bone loss. • Functional analyses of both bone and muscle has been completed and a manuscript has been published in PlosOne: We evaluated changes in bone and muscle of mice exposed to hind limb unloading (suspension) (HLS) unloading alone or in addition to proton and high (H) atomic number (Z) and energy (E) (HZE) (16O) radiation. Adult male C57Bl/6J mice were randomly assigned to six groups: No radiation ± HLS, 50 cGy proton radiation ± HLS, and 50 cGy proton radiation + 10 cGy 16O radiation ± HLS. Radiation alone did not induce bone or muscle loss, whereas HLS alone resulted in both bone and muscle loss. Absolute trabecular and cortical bone volume fraction (BV/TV) was decreased 24% and 6% in HLS-no radiation vs the normally loaded no-radiation group. Trabecular thickness and mineral density also decreased with HLS. For some outcomes, such as BV/TV, trabecular number and tissue mineral density, additional bone loss was observed in the HLS+proton+HZE radiation group compared to HLS alone. In contrast, whereas HLS alone decreased muscle mass (19% gastrocnemius, 35% quadriceps), protein synthesis, and increased proteasome activity, radiation did not exacerbate these catabolic outcomes. Our results suggest that combining simulated space radiation with HLS results in additional bone loss that may not be experienced by muscle. Impact: Our results demonstrate for the first time that exposure to specific simulated space radiation (60cGy of proton+HZE) sensitizes bone to the catabolic effects of simulated microgravity resulting in additional bone loss. Interestingly, this was not the case for unloading-induced muscle loss. As radiation exposure alone did not affect either muscle or bone, countermeasures to mitigate unloading-induced bone and muscle loss may also mitigate the catabolic effects of exposure to space radiation. We are continuing with bioinformatic analyses of bone and muscle.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On the ground, concomitant bone and muscle loss are frequently experienced by individuals confined to bed for long durations, whether a result of neurological injury, trauma, or neuromuscular disease (i.e., cerebral palsy, muscular dystrophy, etc.). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of muscle and bone occurs. In the case of recoverable disease, or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Furthermore, age-related bone and muscle loss is a serious problem in our increasingly aged population. Understanding the interdependence of muscle and bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population, especially those suffering from osteopenia and sarcopenia.
Task Progress:	• Exposed over 90 mice to simulated microgravity and space radiation. • Completed analyses bone and muscle functional data. • Presented four abstracts related to project. • Published two manuscripts. • Isolated RNA from bone and muscle. • Completed RNA-seq on muscle and started bioinformatics analyses.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Krause AR, Speacht TL, Zhang Y, Lang CH, Donahue HJ. "Simulated space radiation sensitizes bone but not muscle to the catabolic effects of mechanical unloading." PLoS One. 2017 Aug 2;12(8):e0182403. eCollection 2017. https://doi.org/10.1371/journal.pone.0182403 ; PubMed PMID: 28767703; PubMed Central PMCID: PMC5540592 , Aug-2017

Awards	Donahue HJ. "City College of New York Sigma Kappa Tau Lectureship, March 2017." Mar-2017
Awards	Donahue HJ. "Orthopaedic Research Society Outstanding Achievement in Mentoring Award, March 2017." Mar-2017